Day 032 | Unit 03 | Teacher plan

Integers and Rational Numbers

Recognize Opposites

Standards: 6.NS.C.6a

Learning target: I can identify opposites and the opposite of an opposite.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Opposites are equally far from zero on opposite sides of a number line. The opposite of 0 is 0. Reversing direction twice returns to the original number.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1

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01 / Learn

Opposites are equally far from zero on opposite sides of a number line. The opposite of 0 is 0. Reversing direction twice returns to the original number.

Worked example 1

Find the opposite of -2. Show both values on a number line.

1. Opposites are equally far from zero on opposite sides of a number line.

2. The opposite of 0 is 0.

3. -2 and 2 are each 2 units from 0 on opposite sides.

4. State the result in the form requested: 2.

Answer: 2

Worked example 2

Select all true statements about -15/2 and its opposite. Explain the role of zero.
A. -15/2 and 15/2 are on the same side of zero.
B. The opposite of the opposite of -15/2 is -15/2.
C. The opposite of 0 is 1.
D. The opposite of -15/2 is 15/2.

1. Opposites are equally far from zero on opposite sides of a number line.

2. An opposite reverses direction without changing distance.

3. Reversing twice returns to -15/2.

4. Zero is its own opposite.

Answer: B, D - The opposite of the opposite of -15/2 is -15/2.; The opposite of -15/2 is 15/2.

Day 032 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the opposite of -3. Show both values on a number line.

02 / Constructed response

Find the opposite of -6. Show both values on a number line.

03 / Constructed response

Find the opposite of -5. Show both values on a number line.

04 / Constructed response

Find the opposite of -4/5 and the opposite of that result. Locate all values on a number line.

05 / Constructed response

Find the opposite of -3/4 and the opposite of that result. Locate all values on a number line.

06 / Constructed response

Find the opposite of -2 and the opposite of that result. Locate all values on a number line.

07 / Constructed response

Find the opposite of -7 and the opposite of that result. Locate all values on a number line.

08 / Multiple select

Select all true statements about -5/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -5/2 is -5/2.
B. The opposite of 0 is 1.
C. -5/2 and 5/2 are on the same side of zero.
D. The opposite of -5/2 is 5/2.

09 / Multiple select

Select all true statements about -7/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -7/2 is -7/2.
B. The opposite of 0 is 1.
C. -7/2 and 7/2 are on the same side of zero.
D. The opposite of -7/2 is 7/2.

10 / Constructed response

Find the opposite of -7/5 and the opposite of that result. Locate all values on a number line.

Day 032 | Unit 03 | Answer key

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1. 3

-3 and 3 are each 3 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

2. 6

-6 and 6 are each 6 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

3. 5

-5 and 5 are each 5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

4. 4/5; opposite again -4/5

-4/5 and 4/5 are each 4/5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

5. 3/4; opposite again -3/4

-3/4 and 3/4 are each 3/4 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

6. 2; opposite again -2

-2 and 2 are each 2 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

7. 7; opposite again -7

-7 and 7 are each 7 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

8. A, D - The opposite of the opposite of -5/2 is -5/2.; The opposite of -5/2 is 5/2.

An opposite reverses direction without changing distance. Reversing twice returns to -5/2. Zero is its own opposite.

6.NS.C.6a | Multiple select

9. A, D - The opposite of the opposite of -7/2 is -7/2.; The opposite of -7/2 is 7/2.

An opposite reverses direction without changing distance. Reversing twice returns to -7/2. Zero is its own opposite.

6.NS.C.6a | Multiple select

10. 7/5; opposite again -7/5

-7/5 and 7/5 are each 7/5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response